Pure Maths 1.1 : Quadratics

Completing the square

$ax^2+bx+c \longrightarrow a(x-k)^2+h$

$\boxed{(h,k) \text{ is the Vertex}}$

if $a \neq 1$, factor out a :

$= a\left(x^2+\frac{b}{a}x\right)+c$

$= a\left(\left(x+\frac{b}{2a}\right)^2-\left(\frac{b}{2a}\right)^2\right)+c$

$= a(x-k)^2+h$

Example:

$y=2x^2+8x+5$

$2(x^2+4x)+5$

$2\left(\left(x+\frac{4}{2}\right)^2-\left(\frac{4}{2}\right)^2\right)+5$

$2(x+2)^2-8+5$

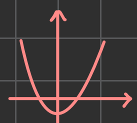
$=2(x+2)^2-3$ #

$[-2,-3]$: Vertex

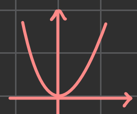
discriminants

discriminant of $ax^2+bx+c=0$ is given by: $[b^2-4ac]$

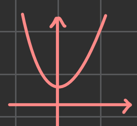
if b^2-4ac is



>0, two real unequal roots



$=0$, one real equal root



<0, no roots, not real

Simultaneous equations

Eg: $y=x-5$ — ①

$5y+x^2+10=-21$ — ②

Substitute eq. ① into eq. ②.

Eg:

$5(x-5)+x^2+10=-21$

~ now solve for x

$5x-25+x^2+10=-21$

$x^2+5x+6=0$

$(x+2)(x+3)=0$

$x=-3$, $x=-2$

Sub x into ① to find y.

Solving Quadratic Equations

Factorizing

Eg: $x^2+5x\underline{-6}$

look for the sum of factors of -6 that equals to the b value, 5.

therefore, roots are $x=-6$ & $x=1$

$\boxed{6x-1 \rightarrow 6-1=5}$

Quadratic Formula

ax^2+bx+c

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Substitute values of abc into formula to find the roots.

Calculator

fx-570ES :

[mode] + [5, EQN] + [3, ax^2+bx+c]

Press values of abc & don't forget negative sign if exists.

Recognizing Quadratic in Other forms & Solving.

equations like

$x^4+5x^2+6=0$

Can be written as a quadratic if you let $x^2=y$, and then :

$(x^2)^2+5x^2+6=0$

$y^2+5y+6=0 \longrightarrow$ Solve for y then, $y=x^2$, find x.

try eq. like $\tan^2\theta+\tan\theta+2=0$ in same.

Pure Maths 1.2 : Functions

Key Concepts & definitions

Function: A rule that assigns each element from the domain (input values) to exactly one element in the range (output values).

Domain: the set of all possible input values (x-values) for a function.

Range: the set of all possible output values (y-values) for a function.

One-One function: a function where each input has a unique output, different x values give diff y-values. This is necessary for function to have inverse.

Inverse function: A function that reverses the effect of original function, denoted as $f^{-1}(x)$. for the function $f(x)$, $f(f^{-1}(x))=x$ & $f^{-1}(f(x))=x$.

Composite function: Combination of two functions, denoted as $gf(x)$ where function f is applied first to x, the output of $f(x)$ is applied into $g(x)$.

Example: $f(x)=x+3$, $g(x)=x^2+x$

$fg(x)$ — g into $f(x)$ so, $fg(x)=(x^2+x)+3$

find value of $fg(3) \longrightarrow fg(3)=(3)^2+3+3$

$=15$ #

Identifying Domain and Range.

Domain: The set values for all x-values for which function is defined, domain can't be 0 if x is denominator of a function or negative in square root.

Range: range of all possible output values eg: for $y=\frac{1}{x-1}$, $y\neq 0$ as $0=\frac{1}{x-1}$ means $\frac{1}{0}=x-1$

undefined.

Composite of function:

a composite function $gf(x)$ is only defined when range of $f(x)$ lies within domain of $g(x)$.

Inverse function range & domain

for inverse functions:

range of $f(x)$ = domain of $f^{-1}(x)$

domain of $f(x)$ = range of $f^{-1}(x)$.

Common domain & ranges

- $\sin\theta$, $\cos\theta$, range $\rightarrow$ $1\geq f(x)\geq -1$
- $\tan\theta$, domain $x\neq 90$ or $\frac{\pi}{2}$ / odd values of $90°$/$\frac{\pi}{2}$.

One-One & Inverse functions

One- One functions

to determine if a function is one-one, check if diff x values always give diff y values. Eg:

$f(x)=x^2$ is not one-one cause $f(-2)=f(2)$.

but $f(x)=x^3$ is one-one.

How to find Inverse of a function

Example: Find inverse of $f(x)=\left(\frac{x-3}{5}\right)\times 6$

#1: Swap the x for y, rearrange eq. to make x subject.

$x=\left(\frac{y-3}{5}\right)\times 6$

$\frac{x}{6}=\frac{y-3}{5}$

$5\left(\frac{x}{6}\right)=y-3$

$5\left(\frac{x}{6}\right)+3=y$

therefore, the inverse $f^{-1}(x)$ is

$$\boxed{f^{-1}(x)=5\left(\frac{x}{6}\right)+3}$$

Graph Transformations

Vertical translation

$y=f(x)+a$, shift up if $a>0$, down if $a<0$.

Horizontal Translation

$y=f(x+a)$ Shift left if $a>0$, right if $a<0$.

Vertical Stretch

$y=af(x)$, Stretch factor a

Horizontal Stretch

$y=f(ax)$, Stretch factor $\frac{1}{a}$

Reflection

In x-axis: $-f(x)$

In y-axis: $f(-x)$

Combined Transformations

Multiple transformations can be applied together:

Example: $y=2f(x+3)-1$

the graph $y=f(x)$ has been

#1: horizontally translated 3 units to left

#2: Vertically stretched by factor 2.

#3: Vertically translated 1 unit down.

* Order of translations:

horizontal shift first, then horizontal stretch, compression, reflect then, vertical stretch, compression, reflect and lastly vertical shift.

Pure Maths 1.3 : Coordinate Geometry

Forms of Line Equations

Slope Intercept form: $y=mx+c$ — m: gradient, C: y-intercept

point-slope form: $y-y_1=m(x-x_1)$ — m: gradient

general form: $ax+by+c=0$

Parallel and Perpendicular Lines

- lines are parallel if their gradient is Same.
- lines are perpendicular if the product of their gradients is -1.

↳ if you have gradient of one line and need to find gradient of its perpendicular Use the formula $-\frac{1}{m_1}$ where m_1 is gradient of original line, $-\frac{1}{m_1}$ is gradient of perpendicular.

Perpendicular bisector of line

$-\frac{1}{m_1}$ = gradient

↳ intersect at midpoint of original line.

#1: find Perpendicular gradient

#2: find midpoint

#3: Use gradient & midpoint in $y-y_1=m(x-x_1)$ to find line eq. of Perpendicular bisector.

Equation of a Straight Line

to form the equation of a straight line you need 2 main things: gradient m and y-intercept c.

$$\text{Gradient }(m)=\frac{y_2-y_1}{x_2-x_1}$$

to find y-intercept (C): $y-y_1=m(x-x_1)$ where y and x remain variable and y_1 & x_1 is a set of coordinate in that line.

Other Formulas:

distance between 2 coordinates

$$=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

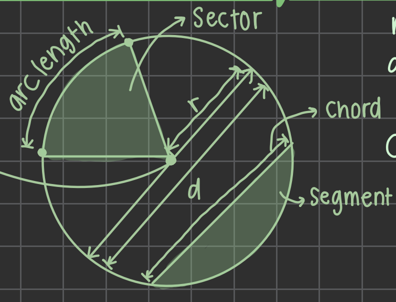
Midpoint of two coordinates

$$=\left(\frac{x_2+x_1}{2},\frac{y_2+y_1}{2}\right)$$

Circles and Their equations

$\pi=3.14$

Centre point of circle.



r: radius

d: diameter

Area $=\pi r^2$

Circumference $=\pi 2r\,/\,\pi d$

Equation of circle

$(x-a)^2+(y-b)^2=r^2$ where (a,b) is centre, r is radius

Note: tangent to circle is perpendicular to radius at point of contact. angle in semicircle is always 90° eg: → always 90°.

Graphs, Intersections and Solving Equations

Intersection points of a line and a curve correspond to the solutions of the simultaneous equations of line and the curve.

Use discriminant (b^2-4ac) to determine how many times a line intersects curve:

$b^2-4ac=0$: tangent (intersects at one point only)

$b^2-4ac>0$: two intersection points

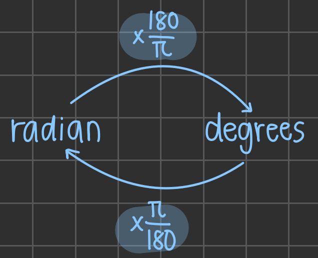
$b^2-4ac<0$: never intersects.

Eg: for $y=x+k$ find number of intersections with given curve by equating them, simplify and use discriminant...

Pure Maths 1.4 : Circular Measure

Conversion of radian to degrees and back

a radian is a unit of angular measure, one radian is the angle opposite to center of circle by an arc whose length is equal to the radius of the circle.



or use ratio

$\boxed{\pi : 180^\circ}$

Arc length, Sector Area & etc

Arc length (s)

$S = r\theta$

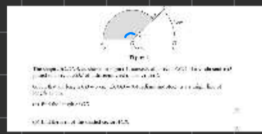
S: arc length, r: radius, θ: angle in radian

Area of a Sector

$A = \frac{1}{2}r^2\theta$

r: radius, θ: angle in radian

Example:



a) arc length CD is 3 cm, θ is 0.4 rad

$S = r\theta$, $3 = r(0.4)$

$r = 7.5\text{cm}$, OD = 7.5 cm

b) Area Sector : $\frac{1}{2}r^2\theta$

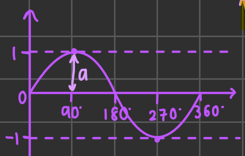
$r = 12 - 7.5$, $\theta = \pi - 0.4$

Area $= \frac{1}{2}(4.5)^2(2.74) = 27.7\text{cm}^2$

Pure Maths 1.5 : Trigonometry.

Graph of Sin x, Cos x and tan x

#1: $y = a\sin bx + c$

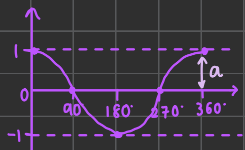


a : amplitude

$\frac{b}{360}$: period

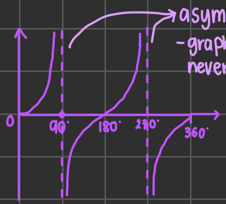
C : Vertical translation

#2: $y = a\cos bx + c$



#3: $y = a\tan bx + c$

no amplitude because it can go up til infinity.



asymptotes
- graph approaches but never meets these lines.

$\frac{b}{180}$ = period

C = vertical translation

a = Vertical Stretch.

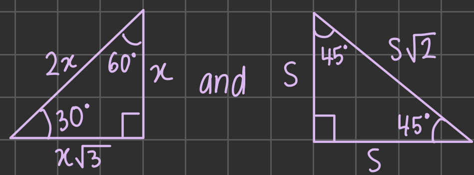
Exact Values of Angles

$\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\tan 30^\circ = \frac{1}{\sqrt{3}}$

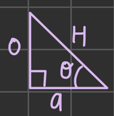
$\sin 45^\circ = \frac{1}{\sqrt{2}}$, $\cos 45^\circ = \frac{1}{\sqrt{2}}$, $\tan 45^\circ = 1$

$\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\tan 60^\circ = \sqrt{3}$

Using special Right Triangles



Note:



$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$ SOH

$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ CAH

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$ TOA

Trigonometric Identities and Solving Equations

① $\sin^2\theta + \cos^2\theta = 1$

② $\frac{\sin\theta}{\cos\theta} = \tan\theta$

③ $\frac{1}{\sin\theta} = \operatorname{cosec}\theta$, $\frac{1}{\cos\theta} = \sec\theta$, $\frac{1}{\tan\theta} = \cot\theta$

S $180-\alpha$ Sin positive only	A all positive α
tan positive only T $180+\alpha$	cos positive only C $360-\alpha$

Using the Identities

Example: $\sec\theta - \sin\theta\tan\theta = \cos\theta$

LHS $= \frac{1}{\cos\theta} - \sin\theta\left(\frac{\sin\theta}{\cos\theta}\right)$

$= \frac{1}{\cos\theta} - \frac{\sin^2\theta}{\cos\theta}$

$= \frac{1-\sin^2\theta}{\cos\theta}$

$\sin^2\theta + \cos^2\theta = 1$
$\cos^2\theta = 1 - \sin^2\theta$

$= \frac{\cos^2\theta}{\cos\theta} = \cos\theta$ = RHS. hence, Proven.

Note:
try changing everything to sin & cos to make Simplifying easier!

For a given range
Example $0 \geq x \geq 360$
u can find all possible values of a trigonometric function using ASTC. Example:

$5\tan^2 x = -16$ don't put negative in calc.

$\tan^2 x = -\frac{16}{5}$

$x^2 = \tan^{-1}\left(-\frac{16}{5}\right)$

α angle

$x^2 = 72.64$

quadrant where T negative: C & S

$\boxed{\begin{matrix}360-\alpha\\180+\alpha\end{matrix}}$

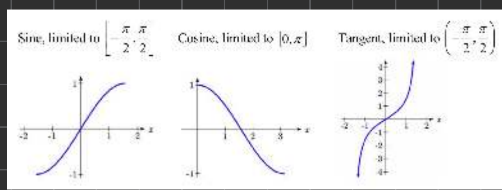
Inverse Trigonometric Function

Inverse of $\sin\theta$, $\tan\theta$, $\cos\theta$ is $\sin^{-1}\theta$, $\cos^{-1}\theta$, and $\tan^{-1}\theta$, which gives the Principle values of angles for specific Intervals. Example:

$\sin^{-1}x$ outputs angle between -90° and 90°

$\cos^{-1}x$ outputs angle between 0° and 180°

$\tan^{-1}x$ outputs angle between -90° and 90°



graphs of Inverse trig functions.

Pure Maths 1.6 : Series

Binomial Expansion

For the expansion of $(a+b)^n$ where n is a positive Integer : Use the binomial theorem.

$(a+b)^n = a^n + {}^nC_1 (a)^{n-1}(b)^1 + {}^nC_2 (a)^{n-2}(b)^2 \ldots$

Until ${}^nC_{n-1}(a)(b)^{n-1} + b^n$.

where ${}^nC_1$ is $\binom{n}{1}$ and is $= \dfrac{n!}{1!(n-1)!}$ Combination formula

Example: $(x+3)^5$

$= x^5 + {}^5C_1 (x)^4(3)^1 + {}^5C_2 (x)^3(3)^2 + {}^5C_3 (x)^2(3)^3 + {}^5C_4 (x)(3)^4 + (3)^5 \longleftarrow$ expand / simplify.

* note : Press [Shift] + [÷] for C in calc.

Geometric Progressions

a sequence where each term is a constant multiple of the previous term. Example:

$2, 4, 8, 64, 128$ ($\times 2$, $\times 2$, $\times 2$, $\times 2$)

common ratio — where $r = 2$

$a = 2$ — first term in Progression.

Formulas

#1 : $T_n = ar^{n-1}$ — n^{th} term formula

#2 : $S_n = \dfrac{a(1-r^n)}{1-r}$ — Sum of n terms when $1 \geq r \geq -1, r \neq 1$

#3 : $S_n = \dfrac{a(r^n-1)}{r-1}$ — Sum of n terms when $r > 1, r < 1$

#4 : $S_\infty = \dfrac{a}{1-r}$ — Sum to infinity only if $|r| < 1$

Arithmetic Progressions

a sequence where the difference between consecutive terms is constant.

Example :

$2, 4, 6, 8, 10$ ($+2$, $+2$, $+2$, $+2$)

d (common difference) $= 2$

a (first term) $= 2$

Formulas

#1 : $T_n = a + (n-1)d$ — n^{th} term

#2 : $S_n = \frac{n}{2}(a + \ell)$ — sum of n terms and ℓ is last term.

#3 : $S_n = \frac{n}{2}(2a + (n-1)d)$ — sum of n terms.

Pure Maths 1.7 : Differentiation

Differentiation Rules

* Power Rule : $\dfrac{d}{dx}(x^n) = nx^{n-1}$
* Chain Rule : $\dfrac{d}{dx}\left(a(x^n-1)^b\right) = ab(x^n-1)^{b-1}(nx^{n-1})$
* Product Rule : $\dfrac{d}{dx}(u(x)v(x)) = u(x)v'(x) + v(x)u'(x)$
* Quotient Rule : $\dfrac{d}{dx}\left(\dfrac{u(x)}{v(x)}\right) = \dfrac{u'(x)v(x) - u(x)v'(x)}{v(x)^2}$

note : Learn indices Rules to help —

$a^m \times a^n = a^{m+n}$ $\quad a^0 = 1$ $\quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$

$a^m \div a^n = a^{m-n}$ $\quad a^1 = a$ $\quad a^{-\frac{m}{n}} = \dfrac{1}{\sqrt[n]{a^m}}$

$(a^m)^n = a^{m \times n}$ $\quad a^{-m} = \dfrac{1}{a^m}$

NOTE :
$u'(x)$ is derivative of $u(x)$
$v'(x)$ is derivate of $v(x)$

Gradient of a Curve

* $\dfrac{dy}{dx}$ represents the gradient of the curve $y = f(x)$

* if value of $\dfrac{dy}{dx}$ at point (x_1, y_1) is m :

→ equation of tangent at that point is $y - y_1 = m(x - x_1)$

→ equation of normal at that point is $y - y_1 = -\dfrac{1}{m}(x - x_1)$

Second derivatives

$\dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) = \dfrac{d^2y}{dx^2}$ or $f'(f'(x)) = f''(x)$.

Example : $\dfrac{d^2}{dx^2}(x^4 + x^2 + 3)$

$\dfrac{dy}{dx} = 4x^3 + 2x$

$\dfrac{d^2y}{dx^2} = 12x^2 + 2$ #

Increasing and decreasing functions

$y = f(x)$ is increasing for given Interval of x if $\dfrac{dy}{dx} > 0$ throughout the Interval.

$y = f(x)$ is decreasing for given Interval of x if $\dfrac{dy}{dx} < 0$ throughout the Interval.

Connected Rates of Change

When two variables, x and y, both vary with a third variable, t, the three variables can be connected using the chain rule :

$$\boxed{\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}}$$

and NOTE :

$$\frac{dx}{dy} = \frac{1}{\left(\dfrac{dy}{dx}\right)}$$

The volume of a spherical balloon is increasing at a rate of 0·05 m^3/s.

How fast is the surface area increasing when the radius is 0·4m?

v = volume, s = surface area, r = radius

$v = \frac{4}{3}\pi r^3 \qquad s = 4\pi r^2$

$\frac{dv}{dr} = 4\pi r^2 \qquad \frac{ds}{dr} = 8\pi r \qquad \frac{dv}{dt} = 0{\cdot}05$

$\frac{ds}{dt} = \frac{ds}{dr} \times \frac{dr}{dv} \times \frac{dv}{dt}$

$= 8\pi r \times \frac{1}{4\pi r^2} \times 0{\cdot}05$

$= \frac{0{\cdot}1}{r}$

When $r = 0{\cdot}4$, $\frac{ds}{dt} = \frac{0{\cdot}1}{0{\cdot}4} = 0{\cdot}25$ Surface area is increasing at $0{\cdot}25 m^2/s$.

Stationary, Minimum and Maximum Points

Stationary (turning points) of a function $y = f(x)$ occurs when $\dfrac{dy}{dx} = 0$

At max/min point, $\dfrac{dy}{dx} = 0$ and the gradient is positive to left of max point and negative to right. gradient is negative to left of min point and positive to the right.

If $\dfrac{dy}{dx} = 0$ and $\dfrac{d^2y}{dx^2} < 0$ it is a maximum point

If $\dfrac{dy}{dx} = 0$ and $\dfrac{d^2y}{dx^2} > 0$ it is a minimum point.

If $\dfrac{dy}{dx} = 0$ and $\dfrac{d^2y}{dx^2} = 0$, nature of stationary point can be found by

Pure Maths 1.8) Integration

Reverse differentiation

* Reverse process of differentiation is Integration

To Integrate $(ax+b)^n$ use:

$$\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

↳ integral symbol ↳ with respect to x ↳ integral constant

Example:

$$\int (2x^2+5x)\, dx = \frac{2}{3}x^3 + \frac{5}{2}x^2 + c$$

* ALWAYS write Integral constant for Indefinite Integrals.

$$\int (2x+1)^2 + 4 \, dx = \frac{\frac{1}{3}(2x+1)^3}{2} + 4x + c$$

$$= \frac{1}{6}(2x+1)^3 + 4x + C$$

Evaluating Integral Constant

* solving for C using given conditions:
- Integrate $\frac{dy}{dx}$ function to find eq. of original line, Substite any known (x,y) coordinates to find C.

EXAMPLE:

$\frac{dy}{dx} = 2x+1$ and curve passes $(1,-2)$ find original curve equation.

$$y = \int (2x+1)\, dx$$

$$y = \frac{2}{2}x^2 + 1x + C$$

$$y = x^2 + x + C \xrightarrow{\text{Substitute}} (1,-2)$$

$$-2 = (1)^2 + (1) + C$$

$$C = -4 \quad \text{so: } [y = x^2 + x - 4] \ \#$$

Definite Integrals

The Integral with limits $[a,b]$

Formula:

$$\int_a^b f(x)\, dx = F(b) - F(a)$$

where $F(x)$ is integral of $f(x)$ without the constant $[+C]$.

Example:

$$\int_1^2 x^2 \, dx$$

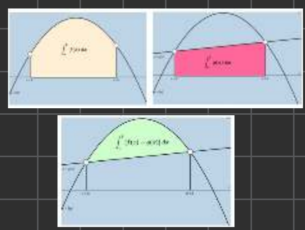
$$= \left[\frac{1}{3}x^3\right]_1^2$$

$$= \frac{1}{3}(2)^3 - \frac{1}{3}(1)^2$$

$$= \frac{7}{3} \ \#$$

Area using Definite Integration

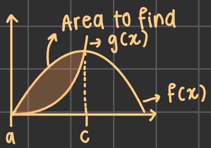
* to find Area under curve, between curve or with respect to lines
↳ Integrate eq. of curve and any other line
↳ Subtract lower curve from upper if needed.



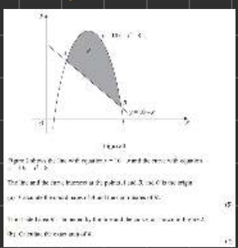
Formula:

$$\text{Area} = \int_a^c [f(x) - g(x)]\, dx$$

where:



Example:



a) A & B are Intersection Points of line $y = 10 - x$ with curve $y = 10x - x^2 - 8$

So, $10 - x = 10x - x^2 - 8$

$x^2 - 11x + 18 = 0$

$(x-9)(x-2) = 0$

$x = 9$, $y = 10 - x = 1$; $x = 2$, $y = 10 - x = 8$

A: $(2, 8)$ B: $(9, 1)$

b) for Area R:

$\int_2^9 (10x - x^2 - 8)$ gives Area:

we don't need area under line $10 - x$

So: $\text{Area } R = \int_2^9 [10x - x^2 - 8] - \int_2^9 [10 - x]$

$$\int_2^9 \left[\frac{11x^2}{2} - \frac{x^3}{3} - 18x\right] = \frac{81}{2} - \left(-\frac{50}{3}\right) = \frac{343}{6} \text{ unit}^2 \ \#.$$

Volumes of Revolution

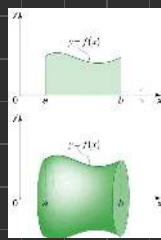
Volume generated by rotating a curve about an axis (y or x).

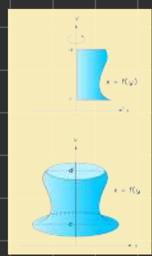
For rotation about x axis:

$$V = \pi \int_a^b [f(x)]^2 \, dx$$

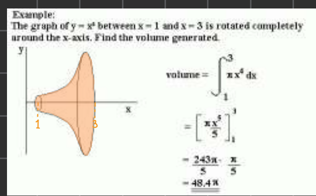
For rotation about y axis:

$$V = \pi \int_a^b [g(y)]^2 \, dy$$





Example:



Example:

The graph of $y = x^2$ between $x = 1$ and $x = 3$ is rotated completely around the x-axis. Find the volume generated.

$$\text{volume} = \int_1^3 \pi x^4 \, dx$$

$$= \left[\frac{\pi x^5}{5}\right]_1^3$$

$$= \frac{243\pi}{5} - \frac{\pi}{5}$$

$$= 48.4\pi$$

extra notes: